

Tester Engineering Suite

v1.0.4-2026 | Operator User Manual

Professional Automotive Diagnostic & ECU Flash Programming

This manual covers installation, licensing, every application tab, the common diagnostic and flash-programming workflows for each supported manufacturer, remote reprogramming, the vehicle history ledger, the accredited emissions lane, and troubleshooting guidance.

Developed by Tester Present Specialist Automotive Solutions



Developed in Australia

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Chapter 1 Introduction & System Requirements

Tester Engineering Suite is a professional automotive diagnostic and ECU flash programming platform targeting workshops, performance tuners, and automotive engineers. It consolidates over twenty manufacturer-specific tools into a single Windows application that talks to the vehicle through any J2534 PassThru adapter.

This manual assumes you are an experienced automotive technician familiar with OBD2 diagnostics, ECU reflashing risks, and J2534 hardware. It is not a tutorial on ECU calibration or a substitute for manufacturer service information.

Supported Operating Systems

- **Minimum:** Windows 7 SP1 (x86 or x64) with .NET Framework 4.8.1
- **Recommended:** Windows 10 or Windows 11 (x64) with 2 GB RAM
- 64-bit Windows strongly preferred for large flash buffers (ZF 6HP26 at 960 KB, BMW MSS70 dual-memory)

Required Hardware

- A J2534-compliant PassThru adapter (Tactrix OpenPort 2.0, AVDI, Drew Technologies CarDAQ, Mongoose Pro, etc.)
- USB 2.0 or newer port
- A stable 12 V vehicle supply during flash operations -- use a workshop battery maintainer
- For accredited emissions testing: a calibrated tailpipe analyser with OBD-II data alignment

Required Permissions

- Local administrator only when running **install.bat** -- the installer creates the Start-menu entry and the desktop shortcut
- Standard user for the application itself -- diagnostic, flash, and emissions operations all run unprivileged
- For NATA accredited emissions mode: an operator account provisioned in the Lab Accreditation dialog

WARNING: Never flash an ECU over a low battery. A vehicle supply below 12.2 V during a write phase can brick the target module. Always use a workshop power supply or active battery maintainer.

Chapter 2 Installation & Licensing

2.1 Installing the Application

1. Extract the release archive to any location on disk.
2. Right-click **install.bat** and choose **Run as administrator**. Follow the prompts -- the installer copies the program files, creates a Start-menu entry, and places a desktop shortcut.
3. Install the vendor driver for your J2534 adapter (Tactrix OpenPort, AVDI, Drew Technologies, etc.) following that vendor's instructions. The adapter must appear in Device Manager without warnings.
4. Launch the program from the desktop shortcut. The first run performs integrity verification of the protection modules; this takes a few seconds.

Note: An internet connection is required for the first run -- the application contacts the Tester Present licence server to validate your machine. Subsequent launches work offline.

2.2 Activating a Licence

On first launch the application displays the Licence Validation screen with a unique **Machine ID** string. Activation is a one-shot, manual exchange:

1. Copy the **Machine ID** shown on the Licence Validation screen.
2. Email the Machine ID to **contact@testerpresent.com.au** along with your invoice reference.
3. You will receive a valid licence key by reply. Paste it into the field on the same screen and click **Activate**.
4. The application restarts automatically once the key is accepted; on next launch it goes straight to the main window.

Note: Licences are bound to a machine ID derived from a stable hardware fingerprint. Transferring to a new workstation requires a new Machine ID and a re-issued key.

2.3 Verifying the Install

- The title bar shows **Tester Engineering Suite v1.0.4-2026**
- The Connection panel at the top of the application shows your J2534 adapter in the drop-down
- Clicking **Connect** with the adapter plugged into a running vehicle establishes a PassThru session without error

Chapter 3 Connecting to a Vehicle

All diagnostic work begins in the Connection panel at the top of the application. The panel is shared across every tab so you can freely switch between, for example, reading DTCs and running a flash without re-establishing the link.

3.1 Basic Connection Flow

1. Key on, engine off (unless your procedure calls for engine running). Seatbelt on, park brake set.
2. Plug the J2534 adapter into the OBD-II port (typically under the dash, driver's side).
3. In the Connection panel, select your adapter from the **Device** drop-down. If the list is empty, re-install the PassThru driver.
4. Click **Connect**. The status indicator turns green.
5. Click **Read Vehicle Info**. The application auto-detects the protocol, identifies the connected modules, and populates the Vehicle Info tab with VIN, Mode 09 data, and UDS identification DIDs.

Note: Protocol selection is fully automatic. The application probes the bus and picks the right protocol (CAN ISO15765, K-Line ISO9141 / KWP2000, J1850, etc.) based on which modules respond. There is no manual protocol picker -- if you need to inspect what responded where, open the **Network Topology** tab after connecting.

Chapter 4 Application Tour — The Eleven Tabs

The main window is organised as a tab strip. Each tab is a self-contained workspace; switching tabs does not disconnect the vehicle.

Tab	Purpose
ECU Identification	VIN decode, Mode 09 vehicle info, UDS \$22 ReadDataByIdentifier for ECU serials, hardware/software part numbers, supplier IDs.
Network Topology	Multi-protocol auto-discovery across CAN + K-Line. Produces a topology map of responding ECUs plus a global DTC scan across every node found.
DTC Memory	OBD-II stored / pending / permanent codes, UDS ReadDTCInformation (\$19) with full status masks, freeze-frame snapshots. Clear codes with a confirmation.
Emissions Tests	I/M readiness LED panel, Mode 05 oxygen-sensor tests, Mode 06 on-board monitor results, Mode 08 on-board test requests. Launches the IM240/RG240 accredited emissions lane.
Dashboard	FPV-style analog gauge cluster. Central RPM, Speed, Coolant, Fuel. Six auxiliary gauges, each right-click configurable to any of the 68 Mode 01 PIDs.
Live Diagnostics	Tabular PID poller with configurable rate. Mixes standard OBD-II, vendor OEM PIDs, and custom user-defined formula PIDs. Logs to CSV on demand.
Signal Analyser	Multi-PID line chart with zoom, pan, seek. Live histogram, min/max/avg/std-dev. Useful for drive-cycle analysis and calibration validation.
Diagnostic Services	18 UDS service sub-tabs: Session Control, Security Access, Read/Write DID, DTC Management, Memory Access, IO Control, Routine Control, ECU Flash, Raw UDS, and protocol-specific screens (KWP2000, McMESS, Ford-9141, PATS).
Data Player	Replay logged drive files at 0.5x to 4x speed with seek bar. Routes recorded PIDs into the Dashboard and Signal Analyser for offline review.
CAN Sniffer	Raw protocol-agnostic frame capture. Include/exclude filters by CAN ID. Transmit custom frames. Frame statistics and frames-per-second live counter.
PassThru Control	Direct J2534 API access: connection attributes, filters, init sequences, voltage pin control. For advanced users and hardware development.

Chapter 5 Reading & Clearing Diagnostic Trouble Codes

5.1 OBD-II Mode \$03 -- Stored Codes

1. Connect to the vehicle (see Chapter 3).
2. Open the **DTC Memory** tab.
3. Click **Read Stored**. The application issues Mode \$03 on the selected protocol and lists every confirmed emissions-related DTC with its SAE description.
4. Double-click any code to view the freeze-frame snapshot (Mode \$02).

5.2 Pending and Permanent Codes

- **Mode \$07** reads pending codes -- faults that have occurred but not yet confirmed over two drive cycles.
- **Mode \$0A** reads permanent codes -- faults that cannot be cleared by a tool until the vehicle completes a readiness run.

5.3 UDS ReadDTCInformation (\$19)

For post-2008 vehicles on CAN, the UDS-level ReadDTCInformation service provides far richer status data than classic Mode \$03. The DTC Memory tab automatically offers this view when the selected ECU supports UDS.

- Sub-function 0x02 -- ReportDTCByStatusMask, returns every DTC matching the tester-supplied status bits
- Sub-function 0x04 -- ReportDTCSnapshotRecordByDTCNumber, per-DTC extended snapshot data
- Sub-function 0x06 -- ReportDTCExtDataRecordByDTCNumber, manufacturer-specific extended data

5.4 Clearing DTCs

1. Click **Clear All**. A confirmation dialog appears.
2. Acknowledge the confirmation. The application issues Mode \$04 (OBD-II) or \$14 (UDS ClearDiagnosticInformation).
3. Re-read stored codes to verify the clear succeeded.

Note: A successful DTC clear is automatically recorded in the Vehicle History ledger against the vehicle's VIN -- you get an audit record without any extra steps. See Chapter 11.

5.5 Global Multi-ECU DTC Scan

From the Network Topology tab, after the topology map has been populated, click **Global DTC Scan**. The application walks every discovered ECU and polls stored + pending + permanent codes from each, presenting a consolidated fault report.

Chapter 6 Live Data, Dashboard & Signal Analyser

6.1 Live Diagnostics Tab

The Live Diagnostics tab is a tabular, high-rate PID poller. It mixes standard OBD-II PIDs, vendor OEM PIDs from the selected database, and any user-defined custom PIDs.

1. In the **PID Library** panel, tick the PIDs you want to read.
2. Pick a poll rate -- **Slow (1 Hz)**, **Normal (5 Hz)**, or **Fast (10 Hz)**.
3. Click **Start**. Values stream into the grid with live min / max / avg.
4. Click **Export CSV** at any time to save the in-memory log.

6.2 Custom PIDs with Formulas

Open **Tools** → **Custom PID Editor**. Each custom PID specifies the service byte, the identifier, the expected response length, and a formula that converts the raw bytes to an engineering value. Formulas are standard infix arithmetic with the operators **+** **-** ***** **/**, parentheses, and the raw byte references **A**, **B**, **C**, **D**.

```
PID 0x22 0xF40C (Engine Oil Temperature)
Length: 1 byte
Formula: A - 40
Units: degC
```

6.3 Dashboard -- FPV-Style Cluster

The Dashboard tab renders a performance-vehicle instrument cluster. Four core gauges (RPM, Speed, Coolant, Fuel) are fixed; six auxiliary gauges are user-configurable.

- Right-click any aux gauge to open the PID picker. All 68 Mode 01 PIDs are offered.
- Gauge ranges auto-compute from the PID definition (min, max, warn, danger).
- The Dashboard accepts data from both Live Diagnostics and Data Player, so recorded drives can be replayed visually.

6.4 Signal Analyser

A multi-channel oscilloscope for CAN data. Up to eight PIDs can plot simultaneously. Mouse wheel zooms; drag pans; the histogram side-panel shows live distribution, percentile bands, and standard deviation.

Chapter 7 ECU Flash Programming — Vendor Workflows

WARNING: ECU flashing is destructive. A failed flash can render a module non-bootable. Always read and save the factory ROM first. Maintain ignition voltage between 12.6 V and 14.2 V throughout the write phase. Do not disturb the OBD cable.

7.1 General Flash Sequence

1. Connect to the vehicle and select **Diagnostic Services** → **ECU Flash**.
2. **Read** the current ROM to a BIN or VBF file. This is your factory backup; keep it forever.
3. Optional: run **Verify Checksums** on the saved file if the platform supports it.
4. Load the replacement file. The application parses and validates the format.
5. If **Auto-correct checksums before write** is enabled, any recomputed checksums are applied to the file at write time, not to the saved copy.
6. Click **Write**. Progress is shown as a percentage with live transfer rate (kB/s) and remaining time.
7. On completion, a post-flash verification and dependency check runs automatically.

7.2 VW / Audi / Skoda / SEAT (Simos)

- Supported ECUs: Simos 8, 10, 12, 12.2, 16, 18, 18.10, 18.41; MQB DQ381, DQ250, Haldex
- File formats: FRF (encrypted), ODX (Open Diagnostic Data Exchange), BIN
- **CBOOT patching** option bypasses RSA signature checks on unlock-capable ECUs
- **Write-without-erase** exploit preserves ECC across calibration-only updates
- Modes: Standard / Calibration Only / Unlock / Full Flash / Re-Lock

7.3 Ford EEC-VI (Spanish Oak)

- Supported vehicles: Focus, Fiesta, Mondeo, Falcon BA/BF/FG, Transit, Ranger, Mustang, F-150 -- 16 module families
- Format: VBF (Volvo/Ford binary format) with SBL (Secondary Bootloader) delivery
- Dual-checksum auto-correction: VID block (\$100C0-\$1013B) + Data checksum (\$10222)
- 300+ pre-calculated module security keys (CAN Generic KeyGenMk1 + GGDS algorithm)

7.4 BMW MS41/42/43/45, MSS70, MSV70

- Dual-memory architecture handled automatically
- **RSA signature recalculation** after calibration edits -- avoids limp-home on modified ROMs

7.5 GM (PcmHammer integration)

- VPW J1850 protocol for 1996-2007 PCMs
- Standard PcmHammer read/write flow with integrated seed/key unlock

7.6 ZF 6HP26 Transmission

- 960 KB full flash, 128 KB calibration-only partition
- Session \$81 with probe reads required before write
- Adaptation data read (rapid mode \$A1/\$A0) + reset with backup/restore

Chapter 8 Security Access & Seed-Key Unlock

Most modern ECUs gate destructive services behind a seed-key challenge (UDS service \$27 SecurityAccess). The application ships with 40+ vendor algorithms plus pre-calculated key tables.

8.1 Using Security Access

1. Open **Diagnostic Services** → **Security Access**.
2. Pick the **Level** -- typically Level 1 (\$01/\$02) for read, Level 3 (\$03/\$04) or higher for programming.
3. Pick the **Algorithm** from the drop-down. The list is filtered to algorithms that match the currently-connected ECU's responses.
4. Click **Unlock**. The application requests a seed, computes the key, sends the key, and reports success or the NRC on failure.

8.2 Algorithm Families

Family	Vendors	Notes
SA2 Bytecode	VW/Audi/Skoda/SEAT	Script interpreter -- full 16-byte seed / key
GGDS	Ford EEC-VI Spanish Oak	CAN Generic algorithm + 300+ module key table
VKeyGen DLL	Daimler / Mercedes-Benz	100+ vendor security DLLs invoked by algorithm name
PcmHammer	GM PCM (VPW)	Integrated library
LFSR	Mazda, Volvo (24/64-bit)	Linear-feedback shift register
SMARTRA	Hyundai / Kia	PIN calculator from immobiliser seed
JLR Pre-calc	Jaguar / Land Rover	311 module groups, 368 security constants
Honda Pre-calc	Honda	1000+ keys, Honda-specific seed handling
PIN Brute-Force	PSA Group (Stellantis)	Bounded-time PIN recovery for VD46/56, CMM, SIM82
UnlockECU Library	Cross-vendor	40+ additional algorithms, loadable at runtime

Note: Unlock attempts are rate-limited by the ECU itself. A wrong key typically locks you out for 10 to 20 seconds; repeated wrong keys can require a full ignition cycle. Do not brute-force algorithms that already have a known solution.

Chapter 9 Instrument Cluster Programming

9.1 FG Falcon (Mk1 / Mk2 / FGX)

The Ford FG Falcon VDO cluster stores VIN, odometer, FPV feature flags, and calibration tables in an internal EEPROM. Mk1 is 512 bytes; Mk2 and FGX are 1024 bytes. The editor handles both layouts including CRC-CCITT checksum recalculation across every protected region.

- Edit VIN (17 characters, auto-uppercase, with copy-regions for Ford-specific block layout)
- Decode/encode odometer (dual-lookup-table subtraction cipher)
- Speedo calibration: 220 km/h, 260 km/h, or 260 km/h FGX
- Tacho calibration: 7,000 RPM or 8,000 RPM
- FPV splash screen, extra needles, gauge type (oil / boost), boost-gauge cal (1 / 1.5 / 2 bar)
- Shift-alert RPM threshold (0--8,000 RPM in 50 RPM steps)
- Police-Mode flag (Ford Police Interceptor configuration)
- +2 km/h digital speedometer override

9.2 FG2 IPC Full Flash

For Mk2 and later clusters the full 256 KB NEC V850 flash can be written. Used to install custom splash images (13 presets shipped or upload your own BMP).

9.3 VAG Clusters (KWP1281)

Legacy VW/Audi K-Line clusters (Golf IV, Passat B5, early A4/A6) are programmed over KWP1281. The application handles the 5-baud init, block sequencing, and module-specific login sequences.

9.4 GM IPC Flash

GM instrument panels on VPW support a PcmHammer-style programming flow for odometer correction and market-region configuration.

Chapter 10 Immobiliser / PATS Key Programming

10.1 BA Falcon PATS Overview

The Ford BA (and related) Passive Anti-Theft System pairs the PCM's encryption key with the BEM (Body Electronics Module). The Immobiliser tab implements the full 7-step parameter reset procedure.

- CAN IDs: **0x6F6** (PCM) / **0x6F8** (BEM), 8-byte frames, 500 ms cycle
- PATS crypto: addition-with-carry, multiply, rotate, XOR, nibble swap
- DIDs: **0xC104** (programmed key count from BEM), **0xC115** (PCM security state)
- Services: **0xB1 01 9C** (initiate), **0xB1 00 0B 20** (start EEC key learn)
- 10-minute polling cycle with automatic state detection (0x33 / 0x55 / 0xAA)
- TesterPresent keepalive runs automatically through the entire procedure

10.2 Parameter Reset Procedure

1. Open **Diagnostic Services** → **Immobiliser / PATS**.
2. Click **Read Key Count**. The BEM returns the current number of programmed keys.
3. Click **Read PCM Security State**. The initial state must be 0x33 (locked).
4. Click **Start Parameter Reset**. The application begins the 7-step procedure, displaying live progress.
5. When prompted (state transitions to 0x55), turn ignition off, wait 10 seconds, then ignition on within the window shown.
6. The state transitions to 0xAA. The application issues **0xB1 00 0B 20** to kick off EEC key learn.
7. Present each key to the ignition cylinder as prompted. The BEM records each acceptance.
8. On success the state returns to 0x33 and the Key Count reflects the newly-programmed set.

WARNING: The full procedure takes up to 10 minutes. Do not disconnect the OBD link, key off, or move the vehicle during the procedure. A broken flow can leave the BEM in an intermediate state that requires dealer-level tooling to recover.

Chapter 11 Vehicle History & Audit Ledger

Every operation performed by the application is recorded against the connected vehicle's VIN in a per-VIN, append-only event store. The ledger is SHA-256 chained so any tampering invalidates subsequent entries.

11.1 What Is Recorded

- DTC scans (Mode \$03, \$07, \$0A, UDS \$19) and DTC clears (Mode \$04, UDS \$14)
- Every flash read and flash write with file hash, file size, ECU ID, duration
- Pre-flash ROM backups captured automatically before every write
- Security unlocks with algorithm name and level
- Coding / adaptation changes
- Service routine executions
- Emissions tests (IM240 / RG240) with pass/fail and measured values

11.2 Viewing History

1. Open **Tools** → **Vehicle History**.
2. Pick a VIN from the list (left panel). The right panel shows the event log in reverse chronological order.
3. Filter by event type, date range, or free-text search.
4. Click any event to expand its detail payload (file hashes, NRC codes, timings, operator ID).

11.3 Exporting a History Report

1. With a VIN selected, click **Export PDF**.
2. Choose the date range and event categories to include.
3. The generated PDF contains: cover page with VIN + vehicle info, chronological event log, integrity hash of the complete chain, operator signature block, and legal footer.

Note: If the VIN is blank or invalid at connection time the system uses the sentinel **_UNKNOWN_** so that filesystem paths remain valid. History entries made against **_UNKNOWN_** can be re-attributed later by editing the event's VIN field.

11.4 Pre-Flash ROM Backups

The **PreFlashBackup** subsystem automatically snapshots the existing ROM immediately before every write operation. Backups live under `%APPDATA%\TesterPresent\Backups\<VIN>\<ECU>_<yyyyMMdd_HHmmss>.bin`. They are retained until you explicitly purge them.

Chapter 12 Emissions Lane (IM240 / RG240, ISO/IEC 17025)

The accredited emissions lane is a separate area of the Emissions Tests tab, gated by operator authentication and lab accreditation configuration. It is designed to produce reports that meet the content requirements of ISO/IEC 17025:2017 §7.8.2.

12.1 Pre-Flight Checklist

- Lab accreditation record configured (scope number, issuing body, expiry)
- Operator logged in via the Emissions login dialog (username + password, audit-logged)
- Reference gas cylinders in date (expiry check enforced)
- Calibration records current (last multi-point QC within validity window)
- NTP time source synchronised (required for defensible timestamps)
- Analyser drift check: zero and span gases within tolerance

12.2 Running a Test

1. Connect the vehicle and open **Emissions Tests** → **IM240 Lane** (or **RG240 Lane** for mass-emissions idle / 2500).
2. Enter the vehicle details -- VIN auto-populates from ECU Identification; fuel type, GVM, ADR category are selected manually.
3. Click **Start Test**. The orchestrator begins the speed trace and data capture.
4. The driver follows the on-screen target trace. Out-of-window deviations flag validity warnings.
5. On completion the system computes pass/fail against ADR limits for the vehicle class.
6. Click **Generate Report**. A structured PDF is produced including: test method ID, results, measurement uncertainty, environmental conditions at time of test, reference-gas traceability, calibration lineage, operator and witness signatures.

12.3 Data Retention

All accredited test records carry a signed 7-year retention floor. The system prevents deletion before that period has elapsed. Post-expiry deletion is manual and audit-logged.

12.4 Calibration & Reference Gases

Tools → **Reference Gas Library** manages cylinder records (supplier, concentration, certificate, expiry). **Tools** → **Calibration Records** holds the calibration and drift history for each analyser channel.

Chapter 13 Remote Reprogramming (Master + Slave + Relay)

The remote reprogramming system tunnels the complete J2534 PassThru API over the internet through a secure cloud WebSocket relay. Three parties: the **master console** (the off-site technician), the **relay server** (stateless WebSocket router with HMAC signing), and the **slave agent** (the workstation at the vehicle, with the adapter plugged in).

13.1 Pairing

1. On the **slave** workstation, launch **TesterPresent.SlaveAgent**. It connects outbound to the relay and displays a 6-character session code.
2. The slave operator reads the code to the master technician over the phone.
3. On the master console, open the **Remote** tab and enter the code. Click **Pair**. The session is extended from 10-minute pending to 60-minute paired.
4. From this point the master's Connection panel drives the slave's J2534 adapter. All tabs work remotely -- DTCs, flash, unlock, emissions, everything.

13.2 Security Properties

- Cryptographically random 6-character session codes (case-sensitive, $\sim 36^6$ keyspace)
- HMAC-SHA256 signing on every slave response -- master verifies before acting
- 30-second heartbeat keep-alive; either side dropping the heartbeat tears the session down
- No VPN required; both endpoints connect outbound, firewall-friendly

13.3 Push-Stream Live Data

Since v1.0.4 the relay uses push-model streaming for live diagnostics. When the master asks to stream a PID list, the slave begins sending updates on its own schedule rather than waiting for per-cycle polling. This reduces round-trip latency and frees the master from polling overhead.

13.4 Remote Flash

Firmware files are chunked to 65 KB and delivered with per-chunk SHA-256 verification. A 12-step UDS flash sequence runs on the slave side; the master watches real-time progress. Abort is one click, with a 600-second timeout ceiling to prevent a lost session from hanging the slave.

Chapter 14 Troubleshooting

Symptom	Fix
J2534 adapter not listed	Re-install the PassThru driver from the vendor. Confirm the device enumerates in Device Manager. Reboot the workstation.
Protocol connect returns NRC 0x11 (Service Not Supported)	Wrong protocol or wrong module address. Use Network Topology to confirm what the target actually speaks.
NRC 0x33 (Security Access Denied)	You have not unlocked to the correct level. Run Security Access first at the level required by the service (typically 3/4 for programming).
NRC 0x78 (Response Pending) repeats	The ECU is busy. This is normal during long routines -- the application waits for completion automatically. If it never clears after 30 s, abort and re-send.
Flash read times out mid-stream	Battery voltage dropped. Check the supply. Re-read from the start; partial reads are discarded.
Flash write aborts with checksum mismatch	Auto-correct before write was disabled, and the file has out-of-date checksums. Enable the toggle or run Verify / Correct Checksums first.
Application refuses to launch after machine change	Hardware fingerprint has shifted -- the licence is bound to the previous machine. Email the new Machine ID from the Licence Validation screen to contact@testerpresent.com.au for a re-issued key.
Remote session drops immediately	Slave agent has no internet or the relay URL is wrong. Check slave logs.
Remote session drops every ~30 s	Heartbeat failing. Typically a stateful firewall. Whitelist WSS to the relay host.
Emissions lane refuses to start	Lab accreditation not configured, operator not logged in, or a reference gas / calibration record is expired.
Vehicle History dialog shows blank VIN list	No entries yet, or the history store path is read-only. Check %APPDATA%\TesterPresent\History .
FG cluster EEPROM write reports bad CRC	Fields outside the protected regions were touched. Reload the EEPROM and edit only the exposed named fields.
PATS procedure stuck at 0x55	Ignition cycle window missed. Abort, wait 2 minutes, and start over.

Log Collection for Support

If you need to raise a support ticket, gather the following and attach:

- The **session log** from **Tools** → **Export Session Log** (captures all traffic for the current run)
- The **protection log** at **%APPDATA%\TesterPresent\protection.log**
- For flash issues: the source file, the saved factory backup, and the post-abort state dump
- For emissions issues: the test session JSON at **%APPDATA%\TesterPresent\Emissions\<session>**

Chapter 15 Glossary & Acronyms

Acronym	Meaning
ADR	Australian Design Rules -- vehicle compliance specification
BEM	Body Electronics Module (Ford) -- holds immobiliser state
CAN	Controller Area Network, ISO 11898
DID	Data IDentifier (UDS)
DTC	Diagnostic Trouble Code
ECU	Electronic Control Unit
EEC-VI	Ford's sixth-generation engine control, a.k.a. Spanish Oak
FRF	Audi/VW encrypted firmware container
GGDS	Ford CAN Generic security algorithm
HMAC	Hash-based Message Authentication Code
IM240	Inspection / Maintenance 240-second drive cycle (emissions)
IPC	Instrument Panel Cluster
ISO 14229	UDS -- Unified Diagnostic Services
ISO 15765	Diagnostic communication over CAN (DoCAN)
J2534	SAE PassThru API -- hardware-agnostic ECU programming
KWP	Keyword Protocol (ISO 14230 / 1281)
LFSR	Linear-Feedback Shift Register
Master / Slave	Remote mode roles -- Master is the technician workstation, Slave is at the vehicle
NATA	National Association of Testing Authorities (Australia)
NRC	Negative Response Code (UDS error)
OBD-II	On-Board Diagnostics II, SAE J1979 / ISO 15031
PATS	Passive Anti-Theft System (Ford immobiliser)
PCM	Powertrain Control Module
PID	Parameter ID
RG240	Restricted-Gas 240-second idle + 2500 rpm procedure
SBL	Secondary Bootloader
SA2	Seed-key Algorithm 2 (VAG bytecode)
TCM	Transmission Control Module
UDS	Unified Diagnostic Services (ISO 14229)
VBF	Volvo/Ford Binary Format firmware file
VIN	Vehicle Identification Number
VPW	Variable Pulse Width (GM J1850)
WSS	WebSocket Secure (RFC 6455 over TLS)

End of Manual

Support & licensing: contact@testerpresent.com.au

Documentation & updates: testerpresent.com.au

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